



UNIVERSITY OF  
BIRMINGHAM

School of Engineering

# **Underwater Habitat: Control, Communication & Docking Systems**

*Concept Design Report*

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## Executive Summary

This report presents the concept design of a fully self sufficient ocean habitat located in Rockfall Trough, in the North Atlantic Ocean West of Scotland, operating at a depth of 200m. It is designed to support a crew of 6 members for extended durations. It functions as an offshore sustainable research platform within the Blue economy framework.

The habitat's structure is engineered to withstand external hydrostatic pressure, environmental conditions and long term saltwater exposure. High strength steel pressure hull forms the primary structural exterior. Buoyancy, anchoring and hydrodynamic loading is addressed through hoop stress, buckling and Morison analysis.

Energy is achieved via the usage of tidal stream generation as the primary source. Turbine with 16 meters diameter provides sufficient output to meet a continuous demand of approximately 150kW. The implementation of a battery storage system ensures resilience during low flow conditions. Microgrid architecture is designed to distribute power to all subsystems of the habitat

The design of the life support is based on closed loop environmental control system that manages oxygen generation, CO<sub>2</sub> scrubbing, desalination, humidity regulation, and controlled hydroponic food supplementation.

The habitat integrates three primary mission modules: Direct Ocean Capture, Underwater Data Centre and Deep Sea Mining. DOC capture up to 10 tonnes of CO<sub>2</sub> per day, using electrochemical processing. UDC operates at 200 kW IT load, utilizing seawater cooling. DSM functions as an operating base for autonomous mining crawlers.

Habitat automation is achieved using centralized SCADA based supervisory control system, it integrated sensors from missions, structure, environment to maintain safe operations. A hybrid underwater optical wireless communication network with acoustic backup ensure reliable data exchange. ROVs are utilized to support structure inspection. environmental monitoring and mission operations.

Through integrated structural, energy, life-support, mission, and automation design, the proposed habitat demonstrates a technically feasible, resilient, and scalable offshore platform capable of sustained human occupation and industrial activity in deep-sea environments.

## Hazard and Operability (HAZOP) Analysis

| ID | Hazard                                                                 | Typical Causes                                             | Consequence                                       | Sev<br>(1–5) | Lik<br>(1–5) | Risk | Mitigation                                                     |
|----|------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------|--------------|--------------|------|----------------------------------------------------------------|
| R1 | Hull breach OR flooding                                                | Corrosion, fatigue crack, impact                           | Loss of habitat, crew fatality, mission loss      | 5            | 2            | 10   | Corrosion protection, compartment isolation, ROV inspection    |
| R2 | Overpressure inside habitat                                            | Faulty regulator, blocked vent                             | Structural overload, rupture                      | 5            | 2            | 10   | Relief valves, pressure sensors, safe shutdown                 |
| R3 | Under pressure OR depressurization                                     | Leak, valve failure                                        | Crew injury, life support failure                 | 5            | 2            | 10   | Leak detection, isolation valves, backup O <sub>2</sub>        |
| R4 | Anchor failure = (drift)                                               | Line break, seabed failure, installation error, landslides | Structural overload, tether damage, loss of comms | 4            | 2            | 8    | Redundant anchors, tension monitoring, drift alarm             |
| R5 | Power blackout                                                         | Generator failure, distribution fault                      | Loss of comms/control/life support                | 5            | 2            | 10   | UPS/batteries, redundant supply                                |
| R6 | Electrical short OR fire                                               | Overloading, material failure, water leaks                 | Fire/toxic smoke, equipment loss                  | 5            | 2            | 10   | Sealed enclosures, breakers, smoke detection, fire suppression |
| R7 | Life support gas imbalance (O <sub>2</sub> low / CO <sub>2</sub> high) | Ventilation issues, gas tank leaks                         | Crew illness, incapacitation, mission abort       | 5            | 2            | 10   | Redundant sensors, backup scrubber, emergency purge            |

|     |                                       |                                                             |                                                            |   |   |    |                                                           |
|-----|---------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|---|---|----|-----------------------------------------------------------|
| R8  | ROV docking collision                 | Current, guidance error, misalignment                       | Damage to dock/ROV, mission delay                          | 3 | 3 | 9  | Funnel guidance, approach limits, docking cameras         |
| R9  | Comms loss (primary link)             | Turbidity, misalignment, failure, attenuation increase      | Loss of telemetry/control, mission delay                   | 3 | 3 | 9  | Backup acoustic link, automatic failover                  |
| R10 | Sensor failure OR false readings      | Drift, calibration loss, damage                             | Wrong control decisions, missed hazards                    | 4 | 3 | 12 | Sensor redundancy, voting logic, calibration plan         |
| R11 | Environmental event (strong currents) | Storm-driven currents                                       | Loads increase, docking difficult                          | 4 | 2 | 8  | Current monitoring, go/no-go thresholds                   |
| R12 | Mining activity disturbance           | Debris, vibration, seabed instability                       | Impact damage, geohazard risk                              | 4 | 2 | 8  | Exclusion zone, monitoring, emergency shutdown            |
| R13 | Food Contamination                    | Microbial growth, storage failure, supply error             | Crew illness, incapacitation, mission abort                | 4 | 2 | 8  | Food safety protocols, sealed storage, water testing      |
| R14 | Cybersecurity Breach                  | Unauthorized access, firmware vulnerability, insider threat | Loss of control systems, false commands, data exfiltration | 4 | 2 | 8  | Network segmentation, access control, intrusion detection |

Table 1: HAZOP Analysis

# Concept of Operations (CONOPS)

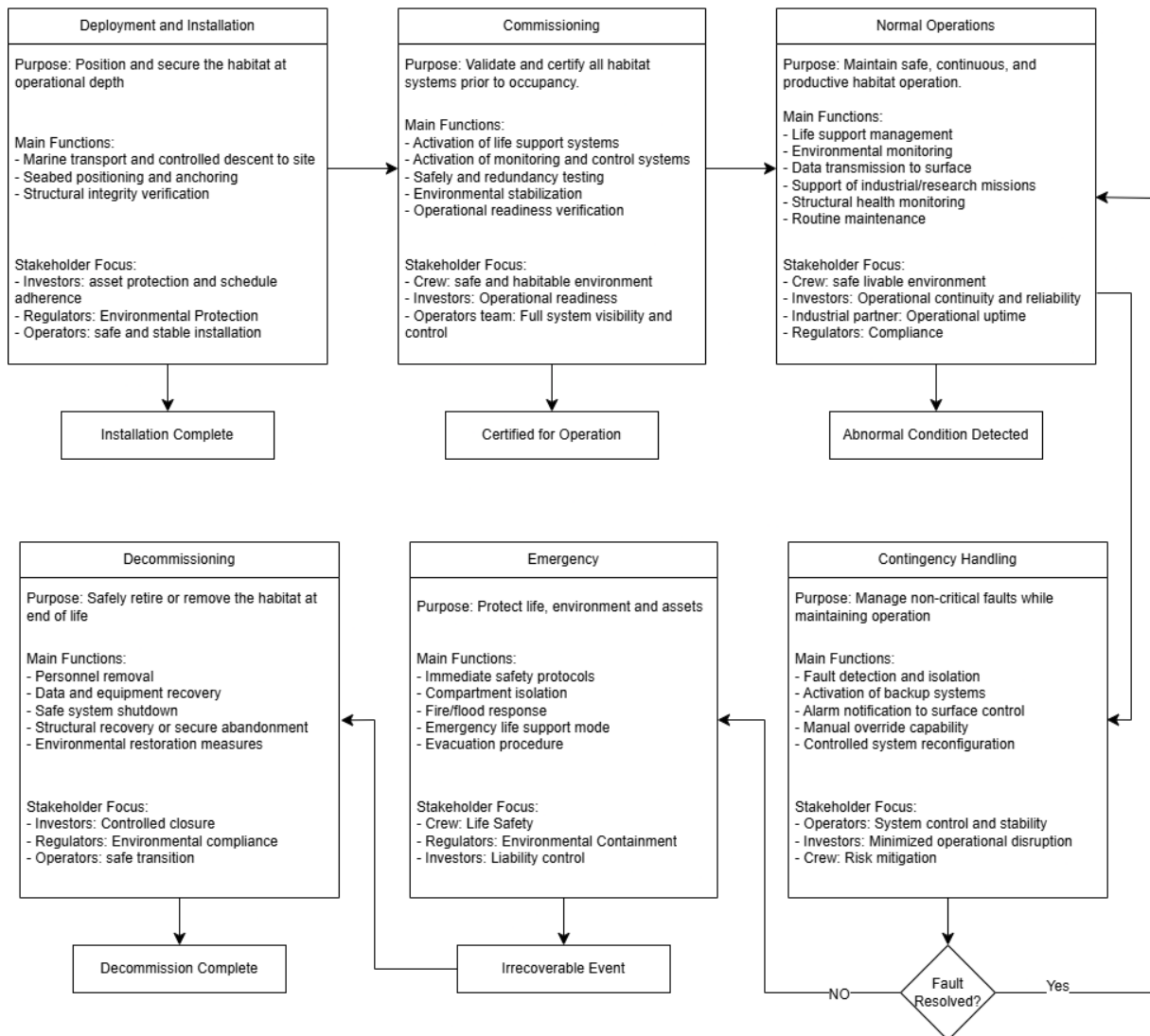


Figure 1: Concept of operations state diagram for the habitat lifecycle

## Site Survey

We have chosen to build our habitat in the Rockall Trough, west of Scotland, in the North Atlantic Ocean at a depth of 200 m. At this depth, the water temperature is approximately 7.9 °C, with salinity ranging from 35.02 to 35.61 PSU. The ambient pressure is roughly 20 bar, which is critical for structural design.

Tidal currents are generally weak offshore, with semi-diurnal tidal currents ranging from 5 to 20 cm/s at this depth. The average tidal range near the shelf edge is 2.5 to 3.0 m, though amplitudes are smaller in the central basin. The seabed consists primarily of mud and silty sediments, which must be considered for anchoring and foundation design.

The Rockall Trough region is exposed to very high waves during Atlantic storms, with significant wave heights up to 18.5 m and extreme individual waves reaching approximately 29 m. These conditions will influence the design of any surface access structures or moorings connected to the habitat.

## Systems Overview

The systems architecture forms as the backbone of the underwater habitat mission, it integrates all subsystems into an operational framework. This maintains coordination, mission completion, adaptability to changing conditions and reliable communication between subsystems.

Communication network is designed to connect systems and all deployed ROVs. Due to this habitat being underwater the architecture had to account underwater propagation constraints therefore systems utilize a hybrid network.

- Primary Link: UOWC
- Backup Link: Acoustic Communications

This approach ensures redundancy, fault tolerances and continuous connectivity. The network topology is design to be modular, allowing scalable expansion via additional seabed nodes and surface buoys. The enables future integration of any new mission without redesigning the core architecture.

Control and Monitoring is a centralized computer operating system based on SCADA architecture that is specifically designed to maintain operations of all subsystems effectively. It collects data from all different sensors, processes real time data, performs trends and anomaly detection. Therefore, the system can identify risk early, outline the level of emergency, level of priority and dispatch direct control command to the relevant subsystems to ensure the situation is under control. The system operates under three defined modes:

- Normal Mode: Standard operations
- Inventory Mode: Resource tracking and logistics
- Emergency Mode: Fault isolation and safety critical response

ROVs are critical in the performance of the habitat therefore implementation of a docking station is necessary to ensure safe recovery, servicing and redeployment. The docking mechanism is designed with reliability and reducing mechanical complexity. The station features a passive funnel shape guide structure that ensures smooth mechanical alignment. Electromagnetic clamps are used to secure the vehicle in place. Once secured the station performs automated charging via inductive power transfers, optical based mission data upload/download and integrated diagnostic check prior to redeployment. This design minimize the active alignment complexity, reduces the failure points and supports long term operation.



# System Requirements

The Following tables list all the specification requirements that need to be fulfilled to initiate operations.

| System                 | Requirement Number | Requirement Type | Description                                                                                                              | Rationale                                                                                             | Traceability to Objective tree                                 |
|------------------------|--------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Control and Monitoring | SYS-01             | Functional       | System shall acquire data from all sensors at defined sampling rates (at latency of < 2 sec)                             | Ensure real time awareness and rapid response                                                         | Maintain stable operation via automated decision making system |
|                        | SYS-02             | Quality          | System shall perform validation checks and calibration to detect drift or corrupted data                                 | Ensure data integrity                                                                                 | Maintain stable operation via automated decision making system |
|                        | SYS-03             | Technical        | Critical life support sensor shall implement a 2 out of 3 redundancy voting to eliminate single point sensor failures    | Reduces false alarms or missed hazards                                                                | Regulate atmospheric composition within safe ranges            |
|                        | SYS-04             | Functional       | System shall detect sensor, actuator or communication faults within 5 sec of occurring                                   | Prevents escalation of damages or hazards                                                             | Maintain stable operation via automated decision making system |
|                        | SYS-05             | Quality          | All sensor data, alarms, control actions and any mode changes must be time stamped and logged                            | Provides traceability and analysis for long term trends                                               | Monitor corrosion rates to prevent early hazards               |
|                        | SYS-06             | Technical        | The Supervisory control layer shall be based on SCADA architecture                                                       | Standardization for maintainability                                                                   | Maintain stable operation via automated decision making system |
|                        | SYS-07             | Functional       | System shall operate in three modes: Normal, Emergency and Inventory                                                     | Ensures correct response behaviours                                                                   | Automated emergency backup and recovery                        |
|                        | SYS-08             | Functional       | System shall automatically transition to Emergency once critical alarm conditions have been detected                     | Ensure rapid mitigation of life critical problems                                                     | Automated emergency backup and recovery                        |
|                        | SYS-09             | Functional       | System shall verify issued commands via acknowledgments for subsystems within 5 sec                                      | Prevents failures and enables closed loop feedback                                                    | Monitor generator and storage operations                       |
|                        | SYS-10             | Functional       | System shall classify alarm into three priority levels: Advisory, Warning, Critical                                      | Ensure critical problem and get immediate intervention while minor issues are appropriately addressed | Isolate and safely shutdown hazard                             |
|                        | SYS-11             | Functional       | In Emergency Mode the system shall prioritize essential loads and functions when required                                | Preserve life critical functions                                                                      | Automated emergency backup and recovery                        |
|                        | SYS-12             | Functional       | System shall isolate hazards within 10 seconds of detecting                                                              | Limits the damages to structure and ensures crew safety                                               | Isolate and safely shutdown hazard                             |
|                        | SYS-13             | Functional       | System shall maintain minimum safe condition during emergency mitigation                                                 | Protects crew survival while the situation is mitigated                                               | Maintain comfortable environment for crew                      |
|                        | SYS-14             | Functional       | In Inventory Mode the system shall monitor resources and calculate an estimate time of depletion                         | Enables proactive planning to avoid any shortages                                                     | Maintain high food and water quality standards                 |
|                        | SYS-15             | Quality          | The control interface shall comply with ISO 11064 principles                                                             | Minimise human error                                                                                  | Maintain comfortable environment for crew                      |
|                        | SYS-16             | Quality          | System shall maintain > 99% operation availability in Normal modes and support maintains intervals of upwards of 5 years | Appropriate for offshore remote sites                                                                 | Maintain stable operation via automated decision making system |
|                        | SYS-17             | Customer         | Operator shall be able to monitor system portable, interpret alarm condition and change any parameters                   | Ensuring safety of onsite crew members                                                                | Maintain stable operation via automated decision making system |

Table 2: Control Systems Requirements Specification

| System                | Requirement Number | Requirement Type | Description                                                                                       | Rationale                                                             | Traceability to Objective tree                                      |
|-----------------------|--------------------|------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
| Communication Network | COM-01             | Functional       | System link shall support a range of 50 meters in oceanic waters                                  | Ensures reliable medium distance due to bandwidth                     | Meet the minimum data rate required                                 |
|                       | COM-02             | Technical        | System shall implement a hybrid UOWC and acoustic architecture using the acoustic links as backup | Improves resilience when degradation or misalignment occurs           | Establish redundant communication routes                            |
|                       | COM-03             | Technical        | The optical transmitter shall operate with the blue green wavelength ideally 488 nm               | Minimises attenuation in water                                        | Meet the minimum data rate required                                 |
|                       | COM-04             | Functional       | The optical link shall maintain a BER < 10 <sup>-6</sup> under nominal operating conditions       | Ensure reliable data transmission                                     | Implement error detection and correction                            |
|                       | COM-05             | Functional       | Receiver shall maintain a minimum required SNR of 15 dB                                           | Ensures sufficient signal quality                                     | Meet the minimum data rate required                                 |
|                       | COM-06             | Technical        | Optical link shall implement PPM modulation                                                       | Power efficient and simple                                            | Meet the minimum data rate required                                 |
|                       | COM-07             | Technical        | Communication system shall implement Forward Error Correction such as Reed-Solomon                | Mitigation of error caused by attenuation and scattering              | Implement error detection and correction                            |
|                       | COM-08             | Functional       | Optical link shall maintain latency of <10 ms                                                     | Real time communication using the fast speeds of optical comms        | Meet the maximum allowable latency                                  |
|                       | COM-09             | Technical        | Internal habitat communication shall utilise fiber optic links supporting > 40 Gbps               | Ensures that internal data exchange is immediate                      | Meet the minimum data rate required                                 |
|                       | COM-10             | Functional       | Underwater communication network shall support a mesh topology using buoys, ROVs and seabed nodes | Enhances resilience and scalability and allows multiple routing paths | Establish redundant communication routes                            |
|                       | COM-11             | Quality          | Communication shall implement action to prevent unauthorised interception                         | Ensure data integrity and security                                    | Enforce Encryption                                                  |
|                       | COM-12             | Quality          | System shall switch from optical to acoustic with 5 sec of failure                                | Prevents any communications loss for critical missions                | Establish redundant communication routes                            |
|                       | COM-13             | Customer         | Operator shall be able to view real time comms status                                             | Helps inform maintenance and operational decisions                    | Ensure reliable data exchange between habitat, ROVs, and subsystems |

Table 3: Communication Network Requirements

| System                    | Requirement Number | Requirement Type | Description                                                                                                                            | Rationale                                                              | Traceability to Objective tree                                           |
|---------------------------|--------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| ROV and Docking Interface | ROB-01             | Functional       | System shall support autonomous ROV homing and docking at the habitat docking station                                                  | Enables safe vehicle recovery                                          | Implement homing system                                                  |
|                           | ROB-02             | Functional       | Docking process shall utilise long range acoustic positioning and short range optical guidance with handover at 25m of docking station | Combines robust long range detection with precise alignment            | Implement homing system                                                  |
|                           | ROB-03             | Functional       | The docking system shall maintain a stable approach under lateral currents up to 0.25 m/s                                              | Ensures reliable docking in ocean currents                             | Account for environmental disturbances                                   |
|                           | ROB-04             | Functional       | Docking interface shall use a passive funnel type guidance system                                                                      | reduces alignment complexity and protects the vehicle while docked     | Ensure secure latching mechanism                                         |
|                           | ROB-05             | Quality          | Power transfer shall use inductive coupling                                                                                            | Improves corrosion and long term reliability due to no relying on pins | Prevent water damage and short circuiting                                |
|                           | ROB-06             | Functional       | Docking interface shall support high power charging of > 500W                                                                          | Minimises charge time and reduces turnaround time for missions         | Ensure quick charge time                                                 |
|                           | ROB-07             | Functional       | AUV shall connect to docking station via optical links to upload data gather upon mission                                              | Enables rapid mission and system diagnostics                           | High upload of mission logs and sensor data                              |
|                           | ROB-08             | Functional       | ROV system shall support inspection of habitat structural integrity, including hull, joints, anchors and subsea modules                | Ensures long term habitat safety                                       | Mission Management                                                       |
|                           | ROB-09             | Functional       | The ROV system shall support environmental monitoring of surrounding water conditions and ecological impact                            | Maintains environmental compliance and sustainability                  | Mission Management                                                       |
|                           | ROB-10             | Customer         | Operators shall be able to monitor docking status, charging progress and control mission                                               | Provides safe human supervision of autonomous systems.                 | Enables multiple safe docking for various ROVs to charge and upload data |

Table 4: Docking System Requirements

## Objective Trees

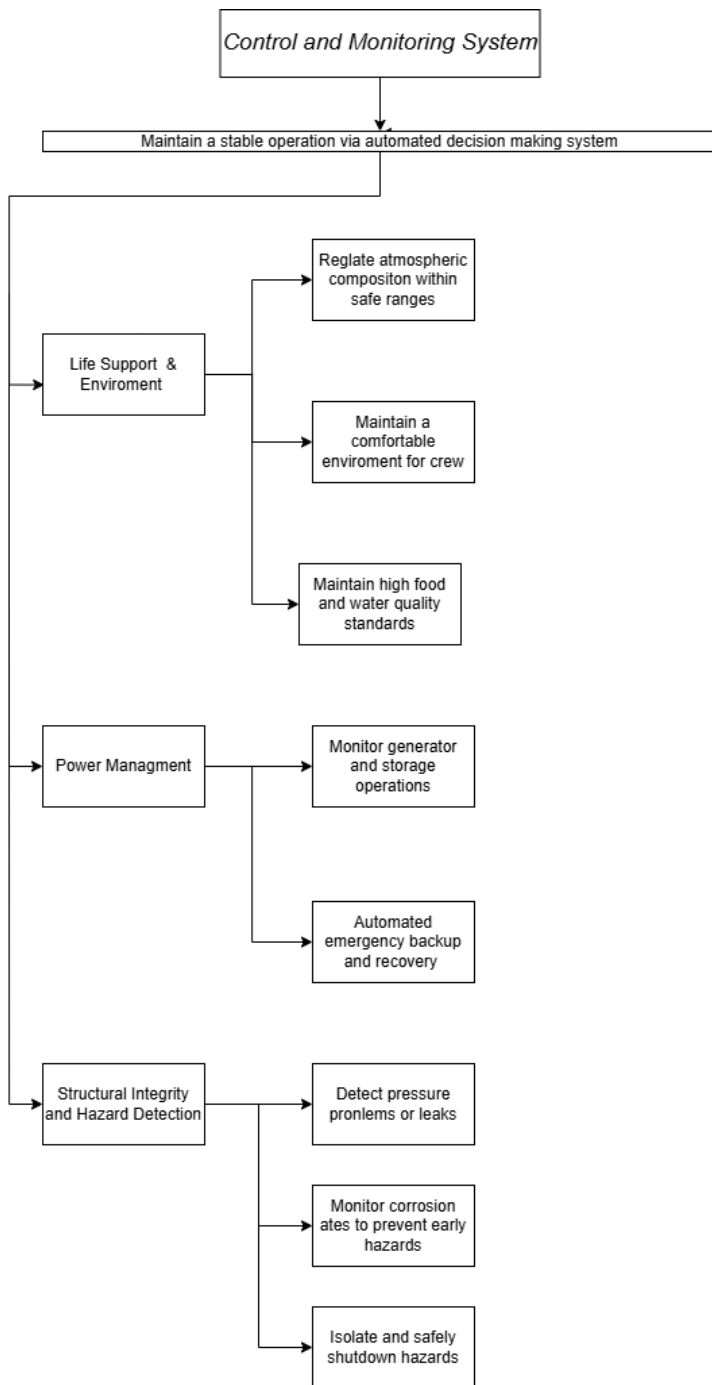


Figure 2: Objective tree for the control and monitoring system

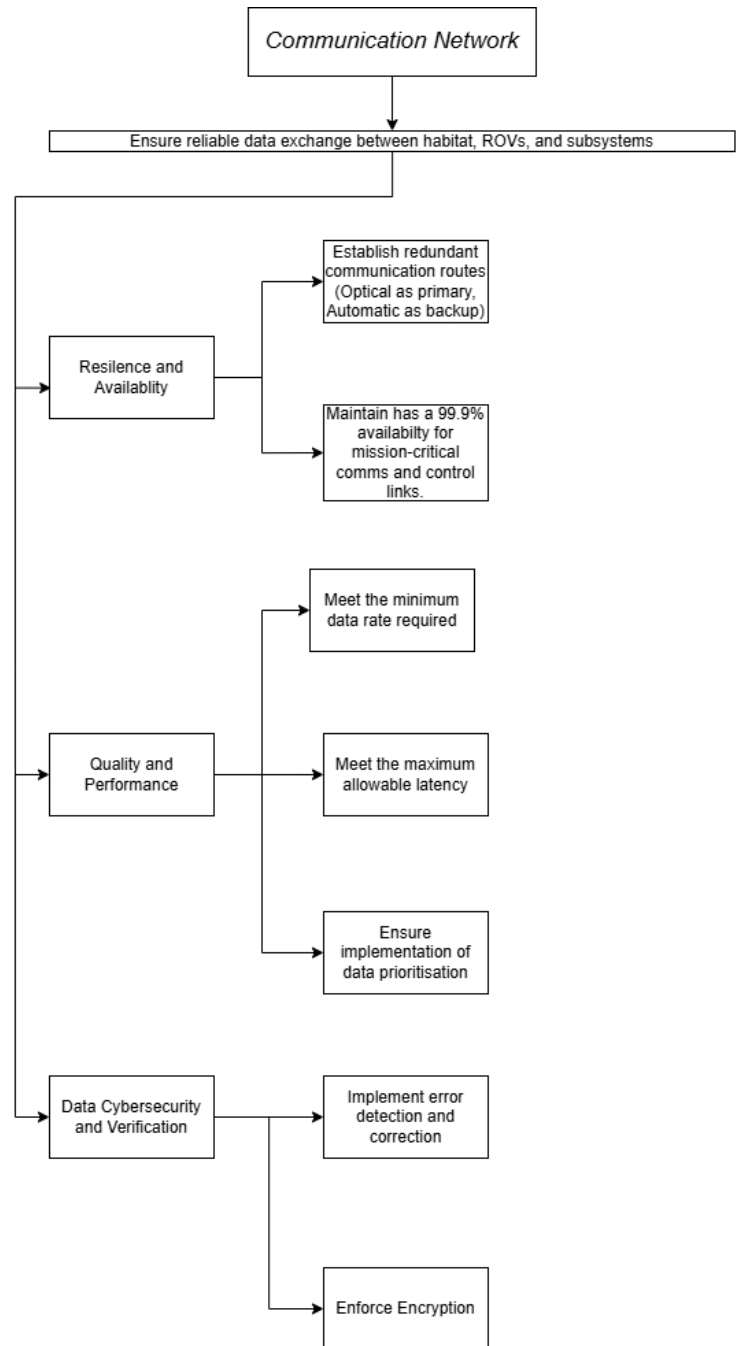


Figure 3: Objective tree for the communication network

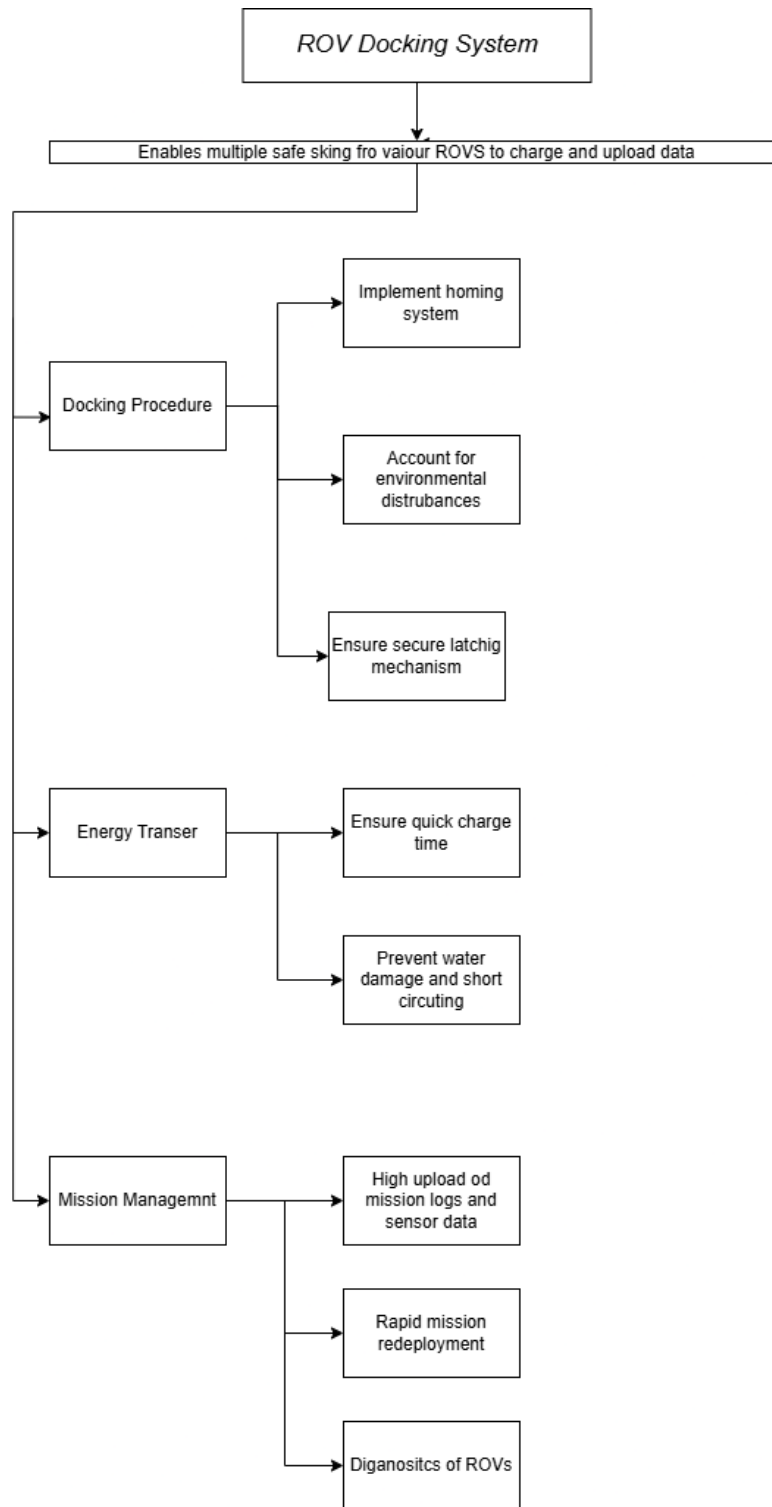


Figure 4: Objective tree for the ROV docking system

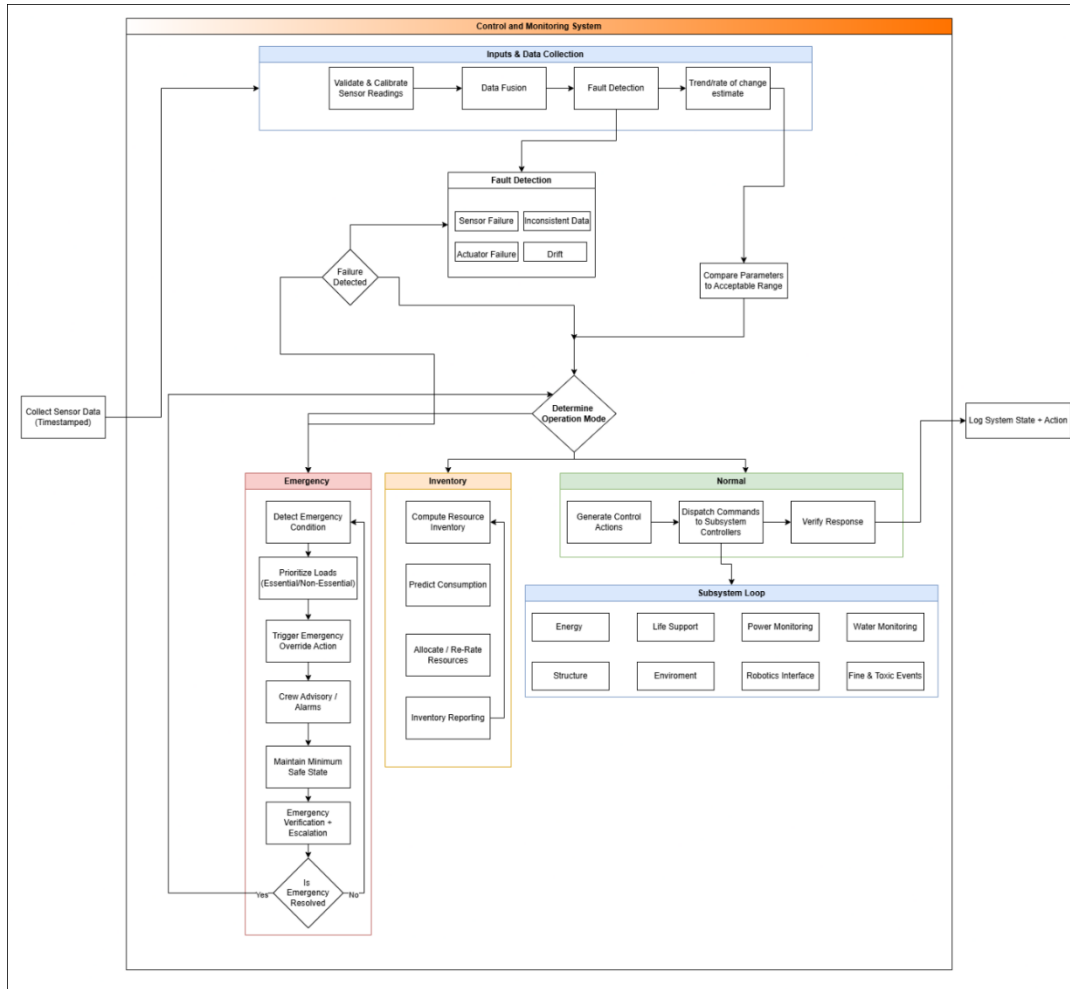


Figure 5: Control and monitoring system process flow

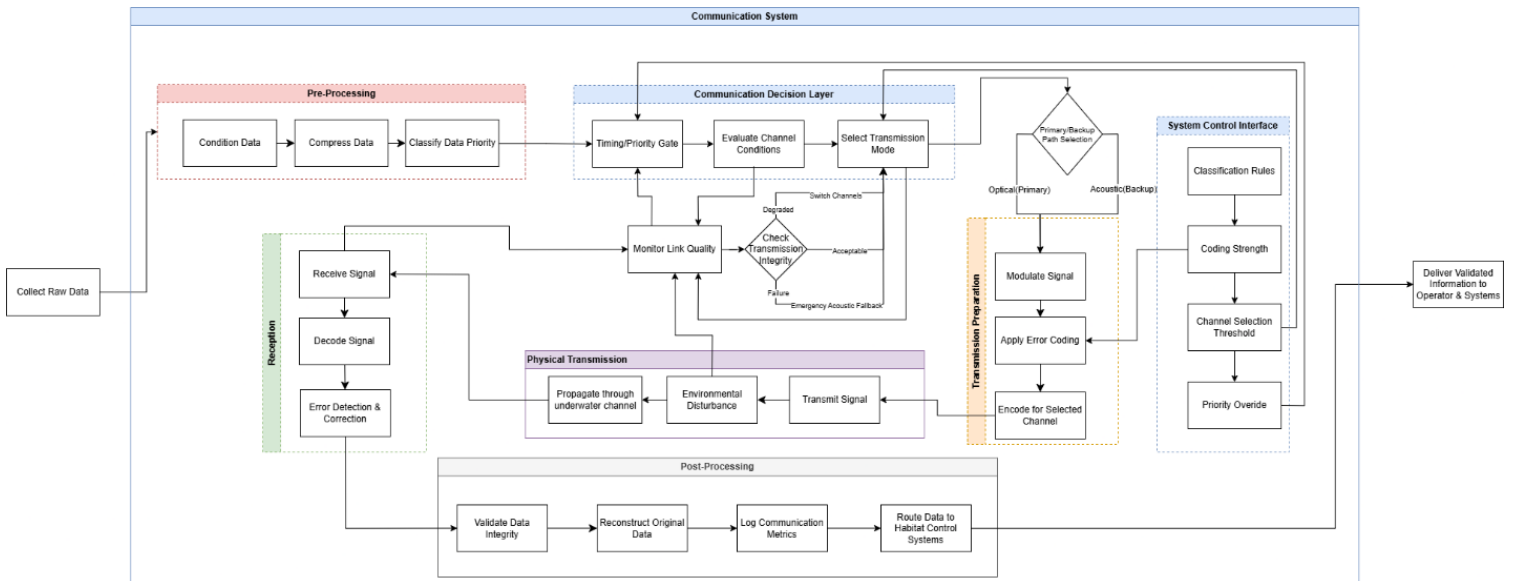


Figure 6: Communication system architecture and data flow

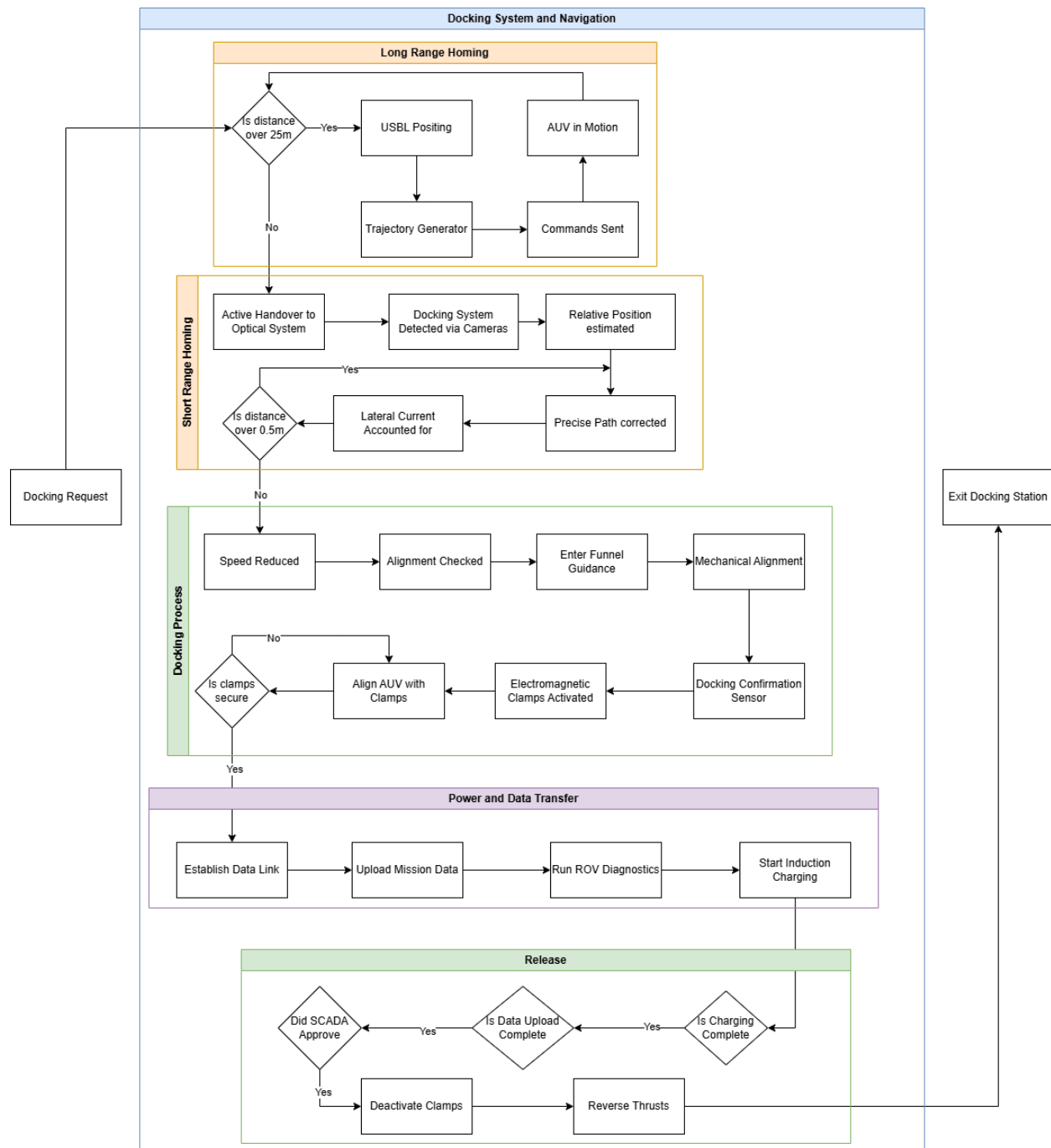


Figure 7: ROV docking and navigation sequence

## **Material Selection: Borosilicate Glass for the UOWC Optical Window**

For UOWC a transparent pressure boundary is required to:

- Transmit 520nm laser light
- Withstand the hydrostatic pressure
- Resist saltwater corrosion
- Maintain dimensional stability
- Avoid thermal cracking

Borosilicate glass was selected due to it fulfilling the criteria.

From the Granta Edupack datasheet we can extract:

### **Mechanical Suitability**

- Young Modules: 49.7 -52.2 GPa
- Tensile strength: 23-25 MPa
- Compressive strength: 232-256 MPa
- Fracture toughness: 0.54-0.55 MPa · m<sup>1/2</sup>
- Density 2100-2150 kg/m<sup>3</sup>

This showcases the ability of adequate compression resistance.

### **Thermal Suitability**

- Thermal expansion coefficient: 3.13–3.9 µstrain/°C
- Thermal conductivity: 1.1–1.2 W/m°C
- Max service temperature: up to 230–430°C
- Excellent thermal shock resistance

Due to low thermal expansion it reduced risk of cracking

### **Optical Suitability**

- Transparent
- Refractive index: 1.46-1.48

This is ideal because it has good transmission and low scattering

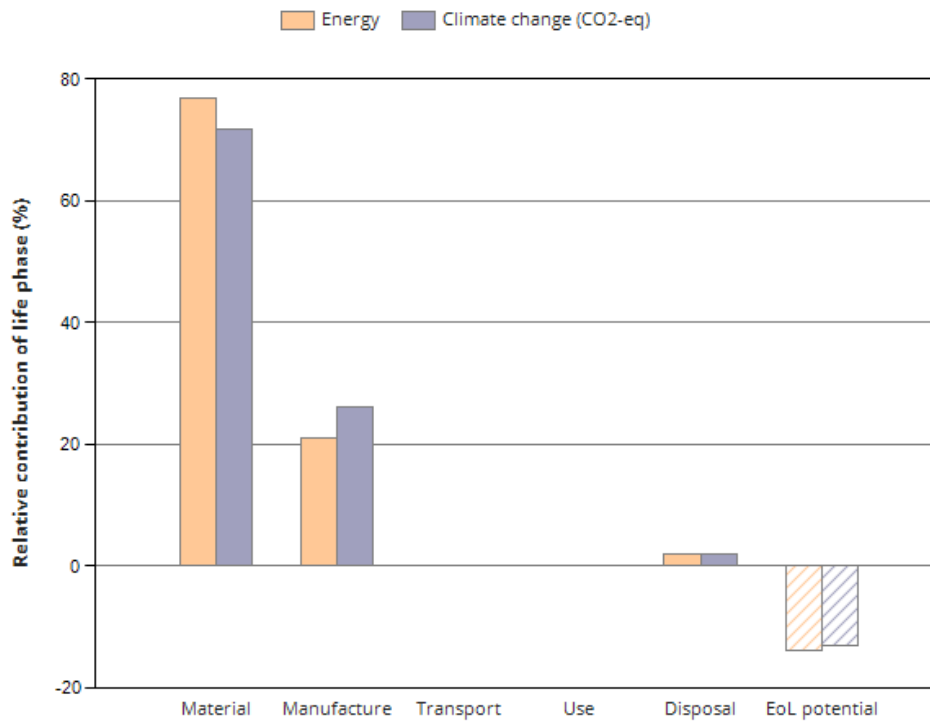
### **Marine Durability**

- Marine atmosphere: Excellent
- Salt water: Acceptable

- Non-flammable
- Non Toxic

## Processing Considerations

- Moldability: 4–5
- Castability: 2–3
- Glass molding energy: 7.5–8.3 MJ/kg
- Recycling fraction: 24%



| Phase                  | Energy (MJ) | Energy (%) | Climate change (CO2-eq) (kg) | Climate change (CO2-eq) (%) |
|------------------------|-------------|------------|------------------------------|-----------------------------|
| Material               | 1.44        | 77.0       | 0.0869                       | 71.9                        |
| Manufacture            | 0.394       | 21.1       | 0.0315                       | 26.1                        |
| Transport              | 0           | 0.0        | 0                            | 0.0                         |
| Use                    | 0           | 0.0        | 0                            | 0.0                         |
| Disposal               | 0.035       | 1.9        | 0.00245                      | 2.0                         |
| Total (for first life) | 1.86        | 100        | 0.121                        | 100                         |
| End of life potential  | -0.26       |            | -0.0157                      |                             |

Figure 8: Eco-audit of borosilicate glass by life-cycle phase



# System Design Philosophy

## Communication Network

### Design Philosophy

The habitat will adopt a hybrid optical- acoustic communication network. UOWC is selected as the primary high bandwidth medium for communication upwards of 50 meters. Acoustic communication provides the ability for communications for long range redundancy and fallback capability. This ensures that the bandwidth performance and the reliability requirements are all satisfied and fulfilled habitat requirements. A mesh network topology is implemented with the usage of both surface level buoys and seabed nodes to expand the coverage.

### Bandwidth Calculations

Optical signal attenuation is modelled using the Beer–Lambert law. For Jerlov Ocean Type II water, the attenuation coefficient is taken as  $K = 0.063 \text{ m}^{-1}$  (Lee et al., 2013). Geometric spreading and hardware inefficiencies are neglected to produce a first-order bandwidth–range estimate.

$$B(z) = B_0 e^{-Kz}$$

Then define:

- $B(z)$ = achievable bandwidth at range  $z$
- $B_0$ = bandwidth at 0 m
- $K = 0.063 \text{ m}^{-1}$

A reference design target of 10 Mbps at 50 m was selected for ROV telemetry and compressed video transmission. This defines  $B_0 = 233.36 \text{ Mbps}$ .

| Range (m) | Bandwidth (Mbps) |
|-----------|------------------|
| 0         | 233.36           |
| 25        | 48.31            |
| 50        | 10               |
| 75        | 2.07             |
| 100       | 0.43             |
| 125       | 0.089            |
| 150       | 0.018            |

The results show exponential bandwidth decay with range. While high data rates are achievable below 50 m, performance rapidly decreases beyond 100 m.

## **Optical Link Configuration**

The system utilized both Point to Point LOS and diffused LOS. Point to point is used to communicate between static nodes that ensure maximum energy efficiency. ROVs and moving nodes will however use diffused LOS to decrease the precision of pointing and misalignment tolerances.

Lasers using the wavelength of around 450nm is chosen because the blue-green region will minimise scattering and absorption, which reduces attenuation. PSK modulation is used to improve the integrity and noise tolerances. OOK modulation is used to reduce the complexity and power consumption for static nodes; ROVs utilise DPIM modulation to improve robustness against misalignment.

Internal communication will use single point fiber optics with an ideal data rate of 40Gbps due to the importance in data logging and communicating with the data center

## **Error Protection and Redundancy**

Reed-Solomon Forward Error Correction is implemented for all life-critical communications. Non-critical telemetry uses lighter error protection to reduce overhead.

## **Control System Architecture**

### **Control Philosophy**

The habitat employs a centralized SCADA supervisory architecture with distributed local controller. The layer provides a global awareness, alarm management and logging capability.

### **Operational Modes**

The system can operate in three different modes to help distribute priorities and tasks

1. Normal Mode: Fully autonomous, optimised for monitoring all subsystems
2. Emergency Mode: Automatically activates upon detection of any critical alarm conditions. It includes load proposition, hazard isolation within 10 secs and maintaining the minimum safety operational state.
3. Inventory Mode: Resource tracking and depletion modelling for life support consumables and operational reserve.

### **Fault Detection and Reliability**

Life support sensor utilize the 2 out of 3 voting redundancy to eliminate any single-point sensor failure that might result in false readings. Fault detection occurs within 5 seconds. Command acknowledgements for subsystems are verified within 5 seconds. All alarms, control actions and mode transitions are timestamped and logged.

The system must be available and should target a 99% working runtime during Normal Operations. Maintenance intervals of at least 5 years.

## **Subsystem Control Loop and Sensor Integration**

The Subsystem loop represents the distributed controllers responsible for closed-loop regulations of major habitat domains:

- Energy
- Structure
- Life Support
- Environment
- Power Monitoring
- Robotics Interface
- Water Monitoring
- Fire and toxic events

Each subsystem operated as an independent control node with health monitoring, fault reporting to SCADA and emergency fallback logic

## **Sensor Layer Integration**

### **Environmental & Structural Monitoring**

- Internal Pressure (Pressure Transducer)
- Water Leak Detection
- Structural Corrosion (ER Probe)
- Vibrations (Triaxial Accelerometer)

### **Life Support Monitoring**

- Oxygen Levels (Electrochemical O<sub>2</sub> Sensor)
- Humidity & Temperature (SHT3x)
- Smoke Detection (MQ-2)
- Thermal Regulation (PT100 RTD)

### **Marine Interaction & External Monitoring**

- CTD (Ocean Quality)
- Hydrophone (Ocean Life Monitoring)

- Tidal Pressure Transducer
- Microbial Food Monitoring (ATP Bioluminescence)

## **Human–Machine Interface**

The Control room is designed using the ISO 11064 design principles. Alarms are classified into Advisor, Warning and Critical levels. The crew members can manually override the system's portability to view and change anything necessary.

## **ROV Docking System**

### **Alignment Strategy**

A funnel shaped passive alignment structure is used to reduce the complexity derived from other docking systems. This helps with alignment tolerances and refitting. ROVs are implemented with USBL if they are over 25 meters from docking station then perform handover operations using optical beacons. Cross-track error based homing control method is used to resist any environmental disturbance that might affect the desired path.

### **Capture Mechanism**

Electromagnetic clamps provide rapid and secure capture without additional actuators or moving parts, thus decreasing the mechanical complexity and failure points.

### **Structural Materials**

The docking structure is constructed from titanium alloy due to its high strength to weight ratio and excellent corrosion resistance in oceanic environments.

### **ROV Connectivity to the Docking System**

Power is delivered using non penetrative inductive charging coils which eliminates any exposed electrical contact, it delivered high speed charging to decrease deployment time. Optical data transfer is activated once secured to upload mission commands and download sensor data from ROV. Vehicle diagnostic are active for evaluation before deployment.

### **ROV Missions and Roles**

The habitat will deploy a fleet of ROVs to help with maintain the underwater habitat and perform specific missions. The following table outlines The ROV types there specific mission role and their equipment

| ROV Unit                                               | Mission Role                                                  | Key Sensors                | Engineering Purpose                                            | Actuators / Equipment           | Design Rationale                                                                 |
|--------------------------------------------------------|---------------------------------------------------------------|----------------------------|----------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------|
| ROV-1 Structural Inspection Unit                       | Inspect hull integrity, joints, anchor points, subsea modules | ER Corrosion Probe         | Detect wall thinning & material degradation                    | 5-DOF Manipulator Arm           | Safety-critical structural monitoring requires high-precision inspection payload |
|                                                        |                                                               | Pressure Transducer        | Detect micro-leaks before catastrophic failure                 |                                 |                                                                                  |
|                                                        |                                                               | Leak Detection Sensor      | Identify internal water ingress                                |                                 |                                                                                  |
|                                                        |                                                               | Triaxial Accelerometer     | Monitor vibration & hull stress under hydrostatic loading      |                                 |                                                                                  |
| ROV-2 Environmental & Sustainability Verification Unit | Monitor ecological impact around habitat                      | CTD Sensor                 | Verify salinity, temperature & turbidity shifts from discharge | Low-noise Thrusters             | Environmental compliance requires low acoustic footprint                         |
|                                                        |                                                               | Hydrophone                 | Detect marine mammal acoustic disruption                       | Peristaltic Water Sampling Pump | Enables water sampling without contamination                                     |
|                                                        |                                                               | Subsea Pressure Transducer | Monitor tidal loads affecting structure                        | Linear Sediment Sampler         | Assess sediment displacement impact                                              |
|                                                        |                                                               | PT100 RTDs                 | Map thermal plume from habitat / UDC                           | Solenoid Release Mechanism      | Controlled sampling deployment                                                   |
| ROV-3 Deep Sea Mining Support Unit                     | Pipeline inspection & equipment servicing                     | CTD (Turbidity Mode)       | Monitor sediment plume during mining                           | Hydraulic Manipulator Arm       | Heavy-duty intervention capability required                                      |
|                                                        |                                                               | Pressure Transducer        | Monitor pipeline integrity                                     | Tool Carousel Motor             | Multi-tool deployment                                                            |
|                                                        |                                                               | Vibration Sensor           | Detect equipment imbalance or failure                          | High-Pressure Pump              | Clearing sediment or assisting pipeline maintenance                              |
|                                                        |                                                               | Corrosion Probe            | Monitor pipeline degradation                                   | Anchoring Claw System           | Stability during intervention                                                    |

The docking station model shown is based on the design principles outlined in *Sparus Docking Station: A Current Aware Docking Station System for a Non-Holonomic AUV* by Esteba et al. (2024).

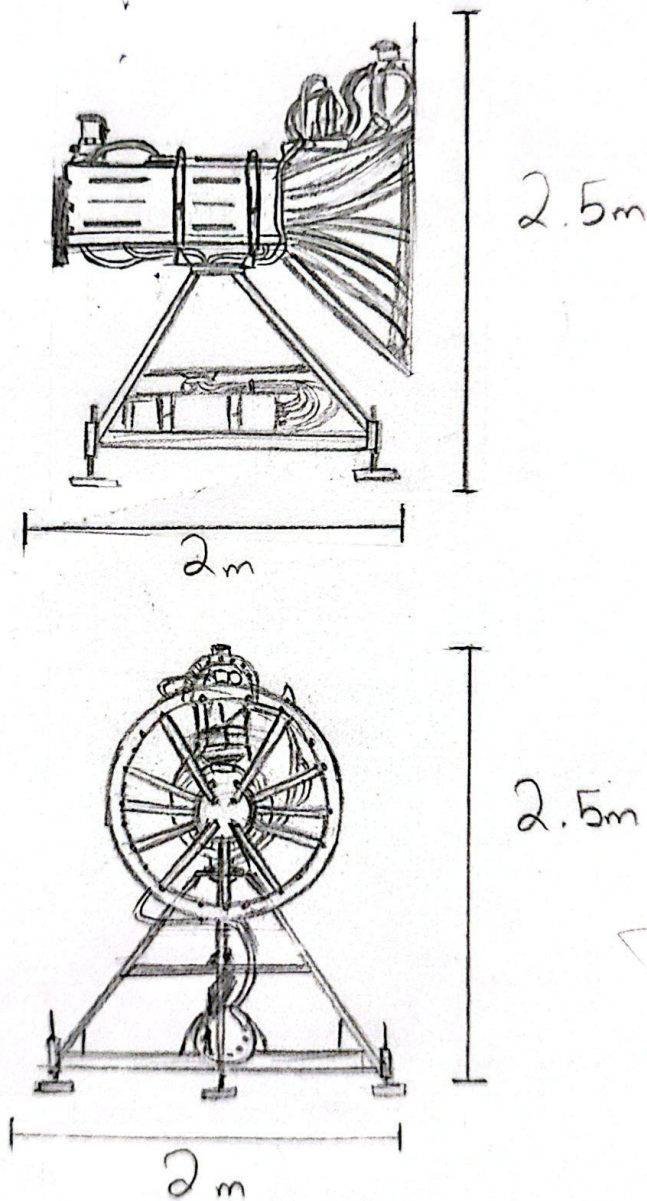


Figure 9: Concept sketch of the docking station

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## Appendix A: Research

### Research Journal Club Reports

**Journal citation:** Zeng, Z., Fu, S., Zhang, H., Dong, Y. and Cheng, J. (2017). A survey of underwater optical wireless communications. *IEEE Communications Surveys & Tutorials*, 19, pp.204–238.  
doi:<https://doi.org/10.1109/COMST.2016.2618841>.

#### Summary:

The paper presents a comprehensive survey of Underwater Optical Wireless Communication (UOWC) as an alternative to acoustic and RF underwater.

It focus on

1. Channel Characterization
2. Modulation Techniques
3. Coding Techniques

UOWC offers significantly higher bandwidth and lower latency than acoustic or RF system but suffers from severe absorption, scattering, alignment sensitivity and environmental turbulence

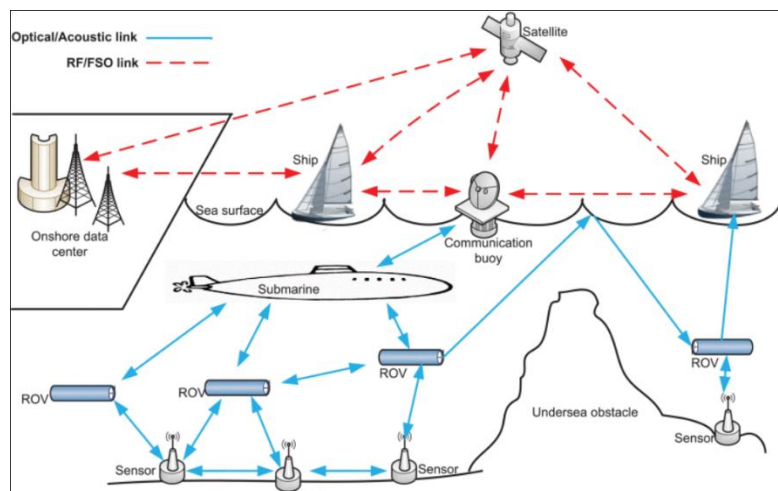
#### Questions (aims):

1. Why should we choose UOWC over others?
2. What are the main physical limitations of UOWC channels?
3. Which link configuration is most suitable for underwater habitat and communication to ROV
4. What channel modulation and coding are most appropriate
5. How can hybrid communication improve system reliability?

#### Answers (knowledge):

1. UOWC is preferred because it fundamentally outperforms acoustic and RF systems in bandwidth and latency.

Acoustic systems can achieve long ranges but their data rate is typically lower with propagation speed. This makes them unsuitable for real time application such as video streaming,



RF underwater comms suffer extreme attenuation in saltwater and limited to very short ranges at low frequencies

In contrast UOWC:

- Operated in the blue green optical widow where seawater attenuation
- Achieve high data rates over moderate distances
- Has negligible propagation delay compared to acoustic systems.
- Provides higher communication security due to narrow beams

For an underwater habitat system where ROV docking requires high speed data upload alongside low latency, optical comms becomes the only viable primary link option

2. UOWC performance is constrained by absorption and scattering,  $c(\lambda)=a(\lambda)+b(\lambda)$

Where:

- $a(\lambda)$ = absorption coefficient
- $b(\lambda)$ = scattering coefficient

Beer lamber law describes first order attenuation:  $I=I_0e^{-cz}$  This shows that received optical intensity decays exponentially with distance.

In pure seawater, absorption dominates, while in coastal or turbid harbor water, scattering becomes severe due to:

- CDOM (Colored Dissolved Organic Matter)
- Phytoplankton
- Detritus particles

Scattering introduces:

- Signal-to-noise ratio (SNR) degradation
- Inter-symbol interference (ISI)
- Timing jitter
- Beam divergence
- Multipath propagation

The blue-green region (~450–550 nm) is least attenuated but even within this window, water type dramatically affects performance.

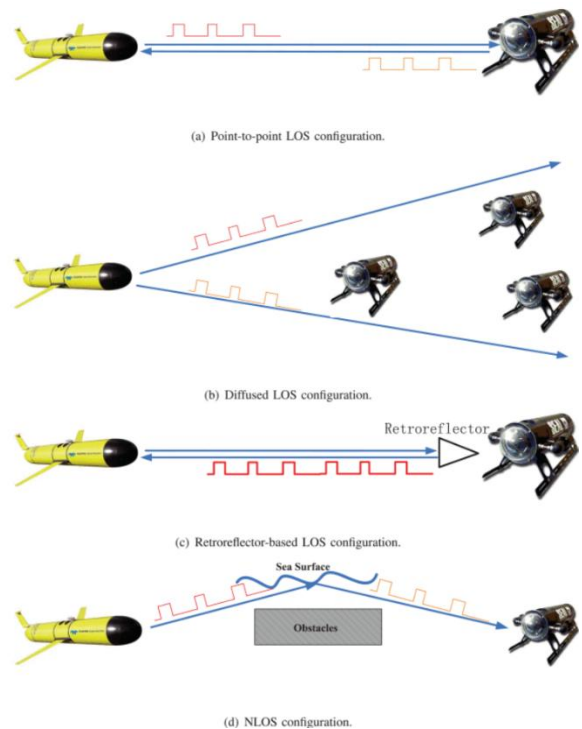
Therefore communication range is not purely a geometric problem by rather it an environmental constraint

3. The paper identified four configs:

- Point to Point
- Diffused
- Retroreflector
- NLOS

For habitat application, Point to point LOS is optimal because it supports high data rates and maximises energy efficiency. Yet it requires high precise alignment which is challenging in turbulent environment such as ROV connections

Therefore using diffused system relaxes the alignment constraints, however it reduces comms range



4. Modulation choices directly affect the power efficiency and bandwidth efficiency.

On-Off keying (OOK)

- Simple
- Binary optical pulses
- Low power and bandwidth efficiency
- Easy implementation
- Suitable for low complexity applications

Pulse Position Modulation (PPM)

- Higher energy efficiency than OOK
- NO dynamic threshold detection needed
- More sensitive to synchronisation jitter
- More complex receiver design

Coherent Modulation (QAM,PSK)

- Higher data rates

- Better BER
- Increase system complexities

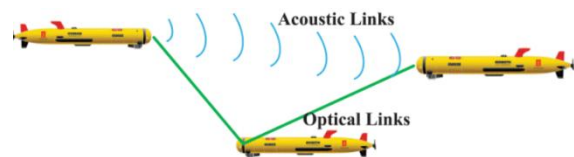
For ROV OOK or PPM is better because of simplicity, energy consumptions and robustness

5. The paper suggests hybrid systems by combining:

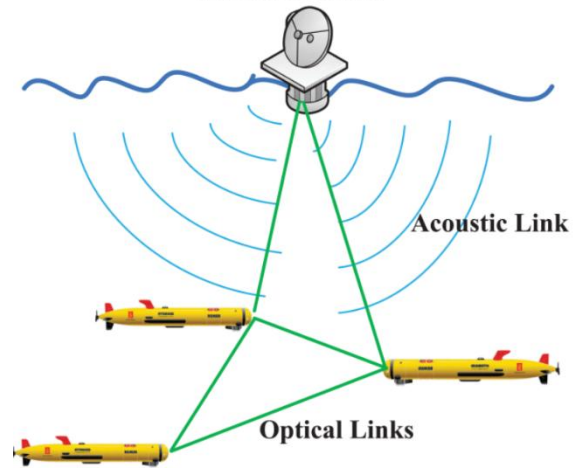
- Optical for short ranges
- Acoustic for long ranges

This layered redundancy increases:

- Fault tolerance
- Mission continuity
- Environmental adaptability



(a) Mutual hybrid UOWC configuration.



(b) Broadcasting hybrid UOWC configuration.

### Impact:

This paper directly informs our habitat communication architecture. It validates the decision to use blue green optical comms as primary high speed link between the habitat and ROVs and acoustic backup as reliability.

It highlights the importance of:

- Accounting for absorption and scattering
- Considering misalignment effects
- Select the right modulation

However we still need to define:

- Internal Habitat connections
- Realistic attenuation coefficient for North Atlantic conditions

### Actions:

- Define exact operating wavelength
- Select modulation scheme
- Perform simplified Beer–Lambert link budget for 50 m and 150 m.
- Design hybrid optical acoustic architecture

**Journal citation:** Jan, Darrell & Seshan, Panchalam & Ganapathi, Gani & Schmidt, Gregory & Doarn, Charles. (1996). Advanced Environmental Monitoring and Control Program: Technology Development Requirements.

**Summary:**

The document establish a structure for EMCS in closed human habitats,

It defines:

- the hierarchical requirements
- the operational mode
- validation
- performance criteria

The document utilises SMAC guideline to establish measurement ranges and sampling frequencies.

**Questions (aims):**

1. Define the EMCS requirements
2. Establish a technology framework

**Answers (knowledge):**

1. The paper outlines the monitoring structural flow:

Mission Definition → System Monitoring Requirements → Species Monitoring → Parameter Monitoring → Technology Assessment.

This ensure that the requirements are mission driven, monitoring is derived from exposure limits, instruments specifications are derived from monitoring needs.

2. Environmental Monitoring and Control System (EMCS) is built to

- Distribute sensors
- A centralised adaptive controller
- Interfaces with life support systems
- Interface with crew
- Interfaces with inventory and logistics
- Fault detection and recognition capability

The controller operates in three modes:

i. Normal Operations:

- Regulates environmental parameters
- Sends commands to life support
- Maintain environmental stability

ii. Emergency Mode:

- Overrides Normal logic
- Takes immediate protective actions
- Can isolate zones and or redirect flows

iii. Inventory Management Mode:

- Tracks Resources storage
- Generated advisory messages
- Interfaces with logistics and resupply

3. The system ensure validation and integrity, by outlining the following rules:

The controller must:

- Validate sensor outputs
- Detect drift
- Detect abnormal trends
- Cross-check data
- Trigger calibration protocols

By including fault detection and predictive models the system can validate the data received and respond accordingly.

**Impact:**

This document is very useful because it outline the control system architecture, even though this is a NASA document regarding extraplanetary activate we can still map it closely to our project due to it being a closed human occupied habitat that is remote and self sufficient, with high risk if failure occurs,

We can adapt

- Mode separation (normal/emergency/inventory)
- Prediction model

- Sensor Validation
- Centralized Control hub
- Resource tracking
- SMACs guidelines

**Actions:**

- Map out the ranges to underwater habitat need alongside the parameters measured

**Journal citation:** Li, D., Chen, Y., Shi, J. and Yang, C. (2015). Autonomous underwater vehicle docking system for cabled ocean observatory network. *Ocean Engineering*, [online] 109, pp.127–134. doi:<https://doi.org/10.1016/j.oceaneng.2015.08.029>.

**Summary:**

This paper showcases a case study of utilising a funnel type autonomous docking system for an AUV that is connected to a cabled ocean observatory network. The system integrates mechanical docking, acoustic and optical navigation, electromagnetic clamping and non penetration inductive power and data transfer.

The docking station must allow the AUV to:

- Detect and localise the docking station
- Navigate toward it under ocean current disturbances
- Be mechanically captured and secured
- Transfer power wirelessly
- Transfer data
- Be safely released

The system uses a USBL acoustic positioning for long range homing and switches to optical vision within 25m for precision docking. Mechanical capture is achieved via three electromagnets (110 N pull force) and the power transfer is done using inductive coupling (500 W across 10mm gap, 80% efficiency). Utilising this funnel design it benefits from being passive, protective and requires minimal refitting.

**Questions (aims):**

1. Why is a funnel-type docking system used and preferred over other active alignment systems?

2. Why combine acoustic and optical positioning instead of using only one?
3. How does the system handle ocean current disturbances?
4. Why is the inductive power transfer is used over a wet mate electrical connectors?

#### Answers (knowledge):

1. The funnel docking system is used because it helps simplify the alignment process from being control algorithmic heavy into a simpler passive mechanical geometry. In sea environments this reliability is much more valuable than mechanical sophistication.

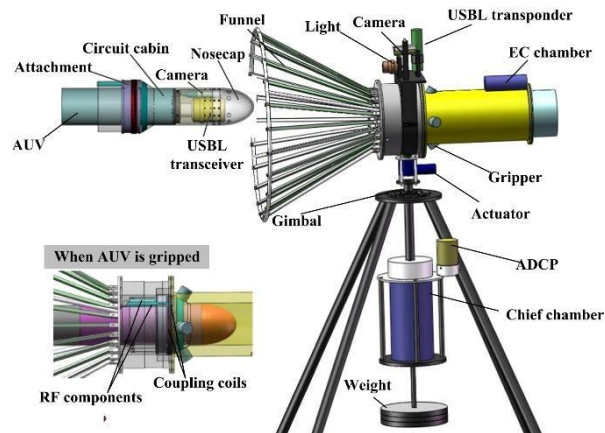
The docking station has a sloped entrance which allows for gradual self centring of the AUV during approach this reduces the need for:

- High precision thruster corrections
- Active robotic alignment arms
- Fine positioning sensors inside the docking tube

By using the geometry to help guide the vehicle in place the system reduces the number of actuators and failure points. In addition once inside the funnel the AUV is shielded from environmental disturbances, improving stability during the final capture.

Compared to other active capture methods, this design is passive, require less retrofitting, mechanically simpler and improves the durability in corrosive subsea conditions

Therefore it makes it well suited for usage in the underwater habitat



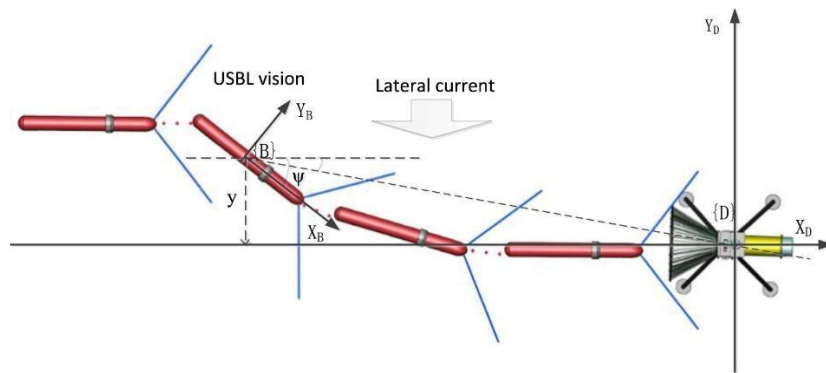
2- Docking is achieved via multiple stages of navigation. When AUV is at long range the homing system utilises USBL acoustic positioning, which measures relative position between the AUV and the docking station. Using Acoustic sensing is very robust over large distances in turbid water but it doesn't have the high precision.

To improve the accuracy at the final approach the system performs a handover sequences which switch the optical vision at a threshold distance of 25 meters. The docking station emits light and the AUV camera performs fine alignment.

This hybrid strategy is used because the acoustic system provides range while optical provides precision. Therefore balancing both constraints.



3- Ocean currents introduce lateral drift that can cause misalignment during docking. The system implements a cross-track-error based homing control method which continuously measures deviations of the desired path and adjust accordingly. The system is capable of resisting the lateral current disturbances up to 0.25m/s, This quantifies tolerance is important because it define the environmental operating limits.



There the control architecture is made up of :

- Closed loop
- Sensor fusion between USBL, gyro and a compass
- Current disturbances

4- After entering the funnel the AUV must be mechanically secured and electrically connected, the system uses three electromagnets that produce a total pulling force of 110N to clamp and secure the vehicle.

Utilising electromagnetic clamping is used because:

- It eliminates mechanical latching there decrease actuators and point of failure
- Engagement and release are electronically controlled.

For the power transfer the station uses inductive coupling power transfer (ICPT)”

- Power transfer capability >500W
- Coil separation gap 10mm
- Efficiency is approximately 80%

Using this non penetration method eliminates exposed electrical connectors, which are highly vulnerable to corrosion and biofouling.

The latching mechanism ensure that the coils are properly aligned which is essential; cause any misalignment would reduce the inductive efficiency.

**Impact:**

This paper is very relevant to the design of the docking system.

Key transferable elements:

- Hybrid acoustic and optical homing architecture
- Funnel base passive docking approach
- Electromagnetic capture mechanism
- Inductive wireless charging system

However the validation was limited to controlled environments and moderate current condition (0.25 m/s). The North Atlantic Ocean has a much stronger current and turbidity so these specific conditions must be considered.

In addition this paper doesn't address:

- How mission data is uploaded/downloaded at the station
- How the docking station performs vehicle health monitoring and self diagnostics while docked

**Actions:**

- Investigate the data upload system
- Define self diagnostics framework
- Evaluate environmental limits for North Atlantic deployment